

Name: _____ Date: _____ Class: _____

MyDesign Portfolio

Element H Initial Template

How to use this template

Guidance before filling out the table on the next page:

- Enter *all* of the high-priority design requirements you listed in Element C into the table below.
- Determine an appropriate testing method for each requirement. For example, the test may involve measurement, inspection, physical actions, or mathematical modeling.
- As you plan, use your scientific method best practices, e.g. creating experimental controls, defining independent and dependent variables, only changing one variable at a time and changing it in increments to find a trend, state your prediction, etc.
- Your testing plan for a given requirement must include enough detail so that a person who is not on the design team could fully replicate the test. Include photos and/or diagrams to add clarity. Each test should include the following details:
 - how the test will be set up
 - what needs to happen during the test
 - test cases (if applicable)
 - number of trials of the test expected (aim for 7 or more trials)
 - the expected range of results that could be recorded, including units
 - the criteria for passing or failing the associated design requirement, including any permissible range for error and/or minimal thresholds
 - any goals your team has set beyond the passing result

Element H: Prototype Testing and Data Collection Plan

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal
1	The prototype must have neutral	1. Place ROV in the water and let PVC pipes fill with water 2. Add and remove pool floaties until the ROV becomes neutrally buoyant 3. Permanently attach pool floaties to the ROV with zip ties	2 trials	Parts of the prototype may: - Completely disconnect from other parts - Sink - Float	Fail: ROV sinks or floats, or completely breaks Pass: ROV is mostly neutrally buoyant Goal: ROV is completely neutrally buoyant

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Element H continues on the next page --

